Examiner reports

Q1.

Question (a) was completed successfully by most students. The most common error was a sign mistake.

Well over half of the students scored at least two marks on part (b) The best students realised they had to do very little to answer this question and, provided they explained their arguments rigorously, they achieved full marks. Most students completed more calculations than necessary (there was no need to find the lengths of all sides), but failed to complete their arguments in sufficient detail. For example, they may have explained why the shape was a parallelogram, but not explained why it was not a rhombus.

Q5.

Again this question was done well by many students although some made minor errors. In part (c) a few students found the acceleration (as $1.4\mathbf{i} + 1.6\mathbf{j}$) and did not calculate the magnitude. Also, in part (d) some students found the angle as 41° rather than 49°. However many students gained full marks on this question.

Q6.

This question was also answered well by the vast majority of students, who showed that they knew the techniques involved in this question. In part (a) the majority of students integrated to find the velocity of the particle but a common error was to forget the + $\mathbf{c}$ term and thus to ignore the initial position vector. Some found the + $\mathbf{c}$ incorrectly, assuming that $\mathbf{c}$ was $6\mathbf{i} - 5e^{-4}\,\mathbf{j}$ without completing the necessary calculation.
Another common error was seen whereby students did not integrate the e^{-4t} term correctly.
In part (b) a number of students found the initial velocity, $-15\mathbf{i} - 5\mathbf{j}$, but did not find the initial speed.

Q7.

This question was also answered well. In part (a), a few candidates did not divide **F** by 50 accurately. In part (b), a number of candidates did not find their '+$\mathbf{c}$' correctly, simply replacing their '$\mathbf{c}$' with $7\mathbf{i} - 4\mathbf{j}$. Others, on evaluating their '$\mathbf{c}$', took $-e^{-2t}$ when $t = 0$ to be +1 rather than –1.

Q8.

There were some vary varied responses to this question. Part (a) was often done well and the candidates who attempted to find the position vector generally gained the marks available. However, quite a few gave the velocity instead of the position vector. Those who had a velocity vector in part (a) then generally used this in part (b) and scored no marks. Some who had a correct position vector in part (a), used the wrong component to try to answer part (b).
Part (c) was altogether more challenging and many candidates worked with position vectors rather than velocity vectors. This clearly was the wrong approach, but some expended a lot of effort solving the quadratic equations that they produced. Some candidates came up with the correct velocity, but without showing how they had obtained their answer. It is important that working is justified. Sometimes a trial and improvement

method led candidates to the correct answer.

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Q10.

Parts (a) and (b) were generally done well, with the candidates gaining many marks. A few candidates could not state the direction in part (b)(ii) and gave answers such as north-east or south-west.Part (c) was found to be more demanding. Candidates who did not identify the **i** and **j** components of the velocity made very little progress. Some made some initial good progress, but made arithmetic or algebraic errors in their solutions. Some forgot to square the speed and tried to work from equations such as $(4 – 0.4t)^2 + (0.5 + 0.2t)^2 = 5$. Some candidates used a trial and improvement approach and realised that the velocity must be $-3\mathbf{i} + 4\mathbf{j}$, which produced a speed of 5.

Q11.

Most candidates answered this question well. In part (a), a small number of candidates forgot to add, and find, the +**c** term and thus ignored the initial position vector; a few candidates tried to use equations for constant acceleration. In part (c), some candidates made errors when finding the force by finding $m\mathbf{a}$, which was $2(6t\mathbf{i} – 8\mathbf{j})$, to be $12t\mathbf{i} – 8\mathbf{j}$.

Q12.

In part (a), many candidates produced correct answers, but some stopped when they had obtained the velocity and did not go on to find the speed as requested. Part (b) was generally done well, but a few candidates did not give their final answer to the nearest degree. Part (c) was very much more demanding. A significant number of candidates mixed scalar and vector quantities in an attempt to answer the question. Statements like

$500 = \frac{1}{2}(0.5\mathbf{i} + 0.375\mathbf{j})t^2$ were quite common and there were some similar ones that also included an initial velocity. The successful candidates used a variety of approaches, which can be seen in the mark scheme. The key difference between those who were successful and those who were not was the ability to connect the scalar distance of 500 to the position vector in a meaningful way.

Q13.

Most candidates answered parts (a), (b) and (c) well, although some did get an answer of

$-4.5\mathbf{j}$ for part (b).

In part (c), the candidates who clearly stated and used a constant acceleration equation usually made a good start and completed the question, although there were a few arithmetic errors.

In part (e) many candidates did not attempt to find the magnitude of the force.

Part (f) was generally found more difficult with only a few good attempts. Several candidates did not attempt this part of the question.

Q14.

Parts (a) and (b) of this question were done very well and many candidates gained all four marks.

Part (c) proved to be more challenging. In part (c)(i), many candidates did not show how they started their working. An equation based on $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ was expected, but not always seen. Some candidates obtained the printed answer, but did not gain full marks as their starting point was not clear. Having both the acceleration and the initial velocity given in the question enabled many candidates to write down a correct expression for the velocity of the particle. Part (c)(iii) was found to be very challenging. If candidates were able to write down an expression for the speed of the particle they would often go on to find the times correctly. It was this vital first step that many candidates were unable to take. A few candidates made errors solving the quadratic equation once they had obtained it correctly.

Q15.

The majority of the candidates were able to find the initial velocity as requested in part (a), but most found the rest of the question difficult. A reasonable number of the candidates were able to find the acceleration, some by calculating two velocities and using $\mathbf{v} = \mathbf{u} + \mathbf{a}t$, but others clearly could not find a strategy to find the acceleration. Part (c) was found to be more demanding. Some candidates did use the components correctly, but quite a number of candidates did not recognise the approach that was required.

Q16.

While some candidates did well with this question, others made some fundamental errors. In part (a), some candidates found the magnitude of the acceleration and then tried to use this scalar quantity in the later parts of the question, ending up with a mix of scalar and vector quantities.

In part (b)(i), many candidates found the position vector, but did not go on to find the distance as requested. There were quite a number of correct responses to part (b)(ii), but quite often this was not followed by a correct approach to the final part of the question.

Q17.

Virtually all candidates answered part (a) correctly. The common problem in part (b) was in the integration of $400\cos\frac{\pi}{2}t$. Apart from this, answers to part (b) were good, and most correctly found $t = 2$ in part (c). The majority of candidates also found correctly that the speed of the particle was 3, with only a few finishing with the velocity of the particle being

–3 or –3**i**.

Q18.

This question also provided more challenge for the candidates. In this question a number of candidates did assume that the acceleration due to gravity was positive.

In part (a), quite a number of candidates found the time of flight and then used half of this value to calculate the maximum height. There were however a number of cases where the candidates did not half their value for the time of flight and so produced solutions which did not relate to the maximum height.

Part (b) was much more demanding and relatively few correct solutions were seen. Those candidates who identified the required method were able to do well, but the majority made no attempt or produced working which could not lead to the required answer. Similarly part (c) was found to quite difficult, but some candidates did use the answer from part (b) to put together correct solutions for part (c) when they had been unable to do part (b).

Q19.

Parts (a) and (b) were generally done well by the candidates. A few candidates subtracted the vectors in part (a) instead of adding them. Also some tried to find the magnitude of each vector in part (a) and then lost several marks because they had very confused working.

In part (c) there were many attempts which involved the division of vectors. For example working such as: $m = \dfrac{9\mathbf{i} + 12\mathbf{j}}{1.5\mathbf{i} + 2\mathbf{j}}$ = 6**i** + 6**j**. Candidates should be advised to consider the components involved and not to attempt to divide vectors.

There were a number of errors that appeared in part (d). One was to include an initial velocity, which was equal to the resultant force on the particle. A few candidates used incorrect formulae. In the final part of the question a large number of candidates gave the vector 3**i** + 4**j** as their final answer and did not find the magnitude in order to get the distance as requested.

Q20.

Most of this question was answered well. In part (a), the only difficulty was found in differentiating $2e^{\frac{1}{2}t}$. In part (b), many found the velocity of the particle to be $(e^{\frac{3}{2}} - 8)\mathbf{i}$ [or –3.52**i**], but did not find the speed of +3.52 m s^{-1}. Others gave the speed as 3.5 m s^{-1}, which was penalised. Those candidates who were successful in part (a) usually completed parts (c) and (d) correctly.

Q21.

This question was also done well by many candidates. In part (a), there were some confused responses and occasional problems with minus signs. Part (b) was also generally done well. In part (c), a few candidates had problems. The most common of these was to omit an arrow from the diagram to show the force; the other was to use the lengths incorrectly when calculating the angle, for example $\tan^{-1}\left(\frac{8}{2}\right)$ and $\sin^{-1}\left(\frac{8}{8.25}\right)$.

Q22.

Generally the candidates made quite good attempts at this question, but some parts caused the candidates more difficulty. A few candidates made poor use of vector notation. Part (a) was answered correctly by the vast majority of the candidates. The most common error was dealing incorrectly with the initial velocity. A few went on to simplify their answers incorrectly. Part (b) was a little more demanding, as some candidates confused the components. Part (c) did cause a little more difficulty for some candidates, with a few candidates not giving any answer at all. However, there were again many correct solutions.

In part (d)(i), although there were many correct solutions, some candidates showed that the helicopter was south or north of the origin, rather than specifically north. Interestingly, some candidates who could not answer part (c) were able to write down the position vector at the specified time. A few candidates found the correct position vector but did not conclude that the helicopter was due north.

When answering part (d)(ii), many of the candidates found the correct velocity, but some did not find the speed as requested.

Q23.

While candidates realised that part (a) required differentiation, a significant number were not able to carry this out accurately. Part (b) proved more successful, but only a small minority of candidates were able to attempt part (c), the simplest solution involving maximisation of the appropriate trigonometrical function.

Q24.

Part (a) was done well, but in part (b) many stopped after finding the velocity at the given time, instead of continuing to find the speed by evaluating the magnitude of the velocity vector.

Q25.

There were very many good solutions to this question and a good number of candidates scored full marks. Some candidates did however make errors. In part (a) there were some cases where the candidates made some minor errors when differentiating one or both of the components. In part (b) a few candidates had difficulties substituting t = 1/3 correctly, but the biggest problem in this part of the question was the description of the direction. While a number did state that the direction was south, some stated that it was north, even though they had calculated the velocity correctly. Others did not use either north or south, but gave answers that implied that **j** was a vertical unit vector.

Parts (c) and (d) were generally done well, but a few candidates experienced difficulties with the differentiation and very occasionally with the substitution of t = 4 .

Q26.

Candidates found this question quite demanding. Candidates were most successful with part (a) and for some this was the only place where they gained marks. In part (b), many candidates omitted the initial position of the yacht, and a number tried to integrate the initial velocity. In part (c)(i), a poor answer for the position vector had an impact. However, some candidates who had good expressions simply showed that the i component of the position was zero and not that the yacht was due north of the origin. In part (c)(ii), quite a few candidates found the velocity correctly, but did not go on to find the speed.

Q27.

There were many good responses to this question. Generally part (a) was done very well. Part (b) was most demanding, but many candidates still did well on this part. One error was to work with a position vector instead of a velocity vector. In part (c), a few candidates gave the velocity as their final answer instead of the speed.

Q28.

Part (a) was frequently done well, with the most common error in part (a)(ii) being the introduction into the equation of motion of an additional force, usually the weight of the motorcycle and rider. Part (b) seldom produced assumptions backed by sound reasons. The majority of responses referred to information given in the question. In part (c), the most convincing solutions were based on the algebraic link between the force and the radius. There was a tendency in part (c) to assume that the magnitude of the frictional force could be altered by changing the radius of the circle, rather than the necessity to adjust the radius to compensate for the force.

Q29.

For those who were well prepared in the calculus techniques linking the quantities involved, this question proved highly successful. Part (c)(ii) proved discriminating, as expected, with only the most able candidates linking the components of the velocity vector with the given vector in an appropriate way.

Q30.

This was a popular starter question often yielding full marks, with only occasional careless errors in arithmetic in part (a) and signs in part (b).

Q31.

It was refreshing to see a wide variety of approaches to this question, which differed in many ways from previous questions on this topic. Many candidates were able to find the coordinates of the ball in part (a) and at this stage an error in the direction of the y-value was condoned. Part (b) provided interesting responses, using either the answers to part (a) with subsequent elimination of the time, or, in some cases, the full equation of the trajectory. In part (c) many candidates used the result of part (b), while others successfully found the time involved as an intermediate step. Part (d) proved to be more discriminating and there was confusion, for some, as to the quantities involved in the motion in the two directions. Many candidates reached the correct result following sound work, but some lost a mark through failing to adhere to the degree of accuracy required by the rubric for the paper. Overall it is encouraging to see candidates who are unable to complete one part of a question continue to other parts where they can be more successful.

Q32.

Most could attempt differentiation correctly to find an expression for the velocity in part (a).

However poor algebra and lack of appreciation of the form of scalar quantities let down many in part (b). Correct answers in part (b) were often then reduced to '6 + 2t' in part (c) and few completed the request correctly.

Q33.

The quality of solutions to problems set on this area of the specification has improved steadily since the first paper in 2001. It was pleasing to see a variety of methods being used and many candidates scored well on this demanding question.

Part (a) was attempted successfully either using the equations of constant acceleration or by integration. The most common error was an incorrect sign in the acceleration of the vertical component.

Part (b) proved to be the most demanding part of the question. Again, varied methods were used, ranging from a full verification of the result to the quotation of the general equation of the path of the trajectory with appropriate substitutions.

Part (c) was popular, yielding marks to most candidates.

The varied methods used in part (d) revealed clear understanding of the motion. The most popular methods used the result quoted in part (b) or the equations of motion with constant acceleration, sometimes incorporating the symmetry of the motion.

In part (e), for those candidates who realised that the greatest speed occurred either at O or at C, the request was straightforward and full marks were often obtained.

Q34.

Parts (a), (b) and (c) were generally answered very well.

In part (a), a few candidates tried to substitute t = 0 into the position vector without differentiating.

In part (c), some candidates found the acceleration, but did not go on to find the magnitude.

Part (d) was more challenging. Quite a few candidates stated that the velocity would become zero. There were some very good answers, but also quite a few that were somewhat confused.

Q35.

The candidates who chose the appropriate constant acceleration equation in part (a) had little difficulty. However, a number did treat the position vector as if it were a velocity vector. In part (b), some candidates stopped when they had found the velocity and did not go on to find the speed as requested. A few candidates had problems dealing with the negative signs. Part (c) was very challenging. Some candidates made good attempts at this, but they were very much in the minority. There were quite a few reasonable answers to part (d).